Feasibility-Aware Quantum-Classical Benchmarking
for Occlusion-Aware UAV Trajectory Optimization

A HUBO formulation with feasibility-first evaluation for quantum optimization workflows

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Scope and novelty. We recast occlusion-aware UAV target tracking as a discrete higher-order binary optimization (HUBO) benchmark and evaluate it with feasibility-aware quantum-classical methodology. Strong classical references are intentional: current quantum hardware limits make honest baselines essential, while the higher-order model creates a path to future D-Wave and QAOA studies.

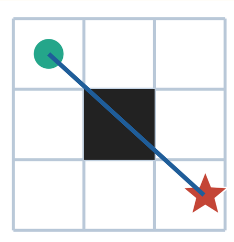
1. Quantum-compatible model
Occlusion, motion, uniqueness, and proximity terms in a single HUBO energy.

2. Feasibility-first metrics
Decode-and-check constraints instead of trusting raw solver energy alone.

3. Baseline-grounded study
A*, SA, neal, and tiny QAOA clarify where quantum methods can matter.

1 Problem

UAV target tracking in cluttered space must choose a collision-free trajectory that also maintains line-of-sight to the target.



- Avoid obstacles and invalid moves.
- Keep the target visible when possible.
- Reach the goal while respecting one-hot position constraints.

Motivation: Masnavi et al. solve the continuous version using multi-convex MPC. We convert the same planning class into a discrete quantum-optimization benchmark.

2 HUBO formulation

Binary state: x[t,v] = 1 iff the UAV occupies grid cell v at time t.

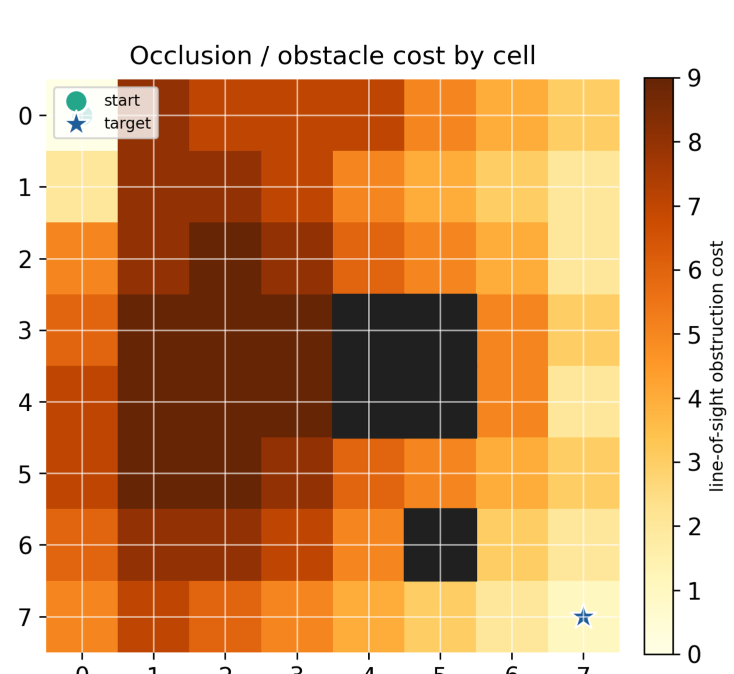
```
H = H_uniq + H_start + H_move + H_obs
    + H_occ + H_prox + H_term
```

Occlusion term
Bresenham line-of-sight checks whether obstacles intersect the UAV-target ray.
H_occ = lambda_occ sum_t sum_u x[t,u] |LOS(u,target) cap O|

Genuinely cubic term
Lingering in a buffer cell is penalized across three adjacent time steps.
H_prox = lambda_prox sum_vb sum_t x[t-1,vb] x[t,vb] x[t+1,vb]

Why HUBO rather than QUBO?
Quadratizing the cubic proximity term would add about **3,172 ancillas** - roughly **30% more variables** - with dense couplings that damage sparsity.

3 Occlusion surface



Yellow cells have higher line-of-sight obstruction cost; dark cells are obstacles. The star marks the target.

4 QCE relevance

- Quantum algorithms and hybrid optimization.
- Benchmarking methodology for constrained quantum models.
- Engineering practice for autonomy, sensing, and planning.

5 Feasibility-aware benchmarking

Raw HUBO energy can mislead. General-purpose annealers may lower energy by violating soft constraints. We therefore separate objective quality from feasibility.

1. **Energy decomposition**
   Report hard penalties and soft costs separately.
2. **Independent constraint check**
   Decode each sample and verify constraints directly.
3. **Matched sampling budget**
   Compare solvers using equal reads/runs.

Key idea: A feasible trajectory with slightly higher objective cost is preferable to an infeasible low-energy sample.

6 Two-tier study design

State-vector QAOA scales as **2^n**, so a full **n = 960** instance is not simulatable on ordinary hardware. We evaluate at two scales:

TIER 1: Full 8 x 8 grid
960 variables
A*, simulated annealing, neal

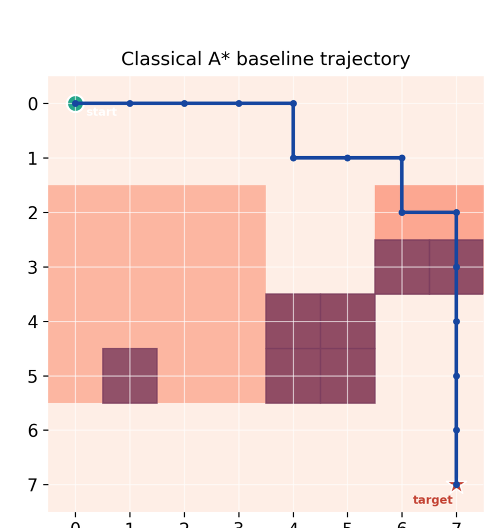
TIER 2: Tiny 2 x 2 grid
8 variables
+ Qiskit QAOA correctness check

Interpretation: Tier 1 asks whether the formulation behaves sensibly at planning scale. Tier 2 asks whether the quantum encoding reaches the same optimum when exact simulation is possible.

7 Classical A* baseline

A* runs on the same grid with an occlusion-weighted cell cost, grounding the HUBO against a conventional planner rather than a benchmark artifact.

```
cell cost = 1.0 + lambda_occ |LOS cap O| +
            lambda_prox 1[v in buffer]
```



A* path: feasible, reaches target, and maintains 80% visibility.

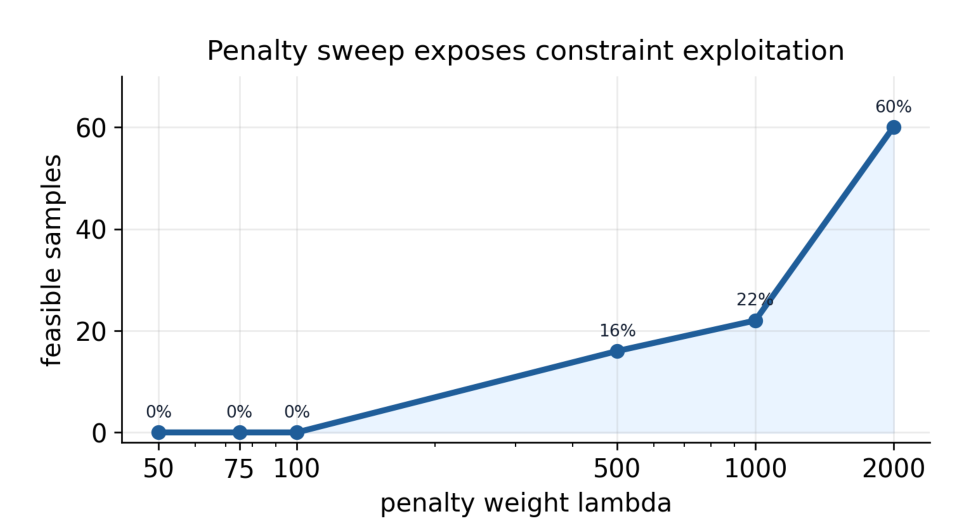
8 Results: 8 x 8 instance

The same problem instance is used across A*, SA, and neal. Feasibility is checked independently from the solver energy.

Metric	A*	SA	neal
Soft cost	317	303	496
Feasible	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	-	-
Runtime / sample	10 ms	47.8 s	0.26 s

Headline: A* and SA reach comparable quality. A* is about **4,700x faster** than SA, while neal is fast but frequently infeasible.

9 Constraint-exploitation curve



neal feasibility climbs from 0% to 60% as lambda increases from 50 to 2000. A single penalty value would misreport feasibility.

10 Takeaways

Formulation value
The HUBO is valuable as a quantum-compatible benchmark, not as a speed win on current hardware.

Fair comparison
Penalty sweeps and direct constraint checks prevent false quantum/annealing advantages.

Quantum sanity check
On 8 variables, all four solvers reach the same optimum (E* = -118.17); QAOA is correct but 2,750x slower.

11 Next steps

Future work: moving targets, real D-Wave hardware, QAOA variants beyond state-vector simulation, and comparison against continuous MPC.

References: Masnavi et al., continuous multi-convex MPC; Qiskit QAOA; D-Wave Ocean / neal. Full citations are in the extended

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